H0A.86 **Wire breakage detection filter time**

Hexadecimal:	200A-57h	Effective Time:	Real time
Min.:	5	Unit:	ms
Max.:	1000	Data Type:	UInt16
Default:	30	Change:	At stop

Value Range:

5 ms to 1000 ms

Description

-

4.12 H0B Display Parameters

H0b.00 **Motor speed actual value**

Hexadecimal:	200b-01h	Effective Time:	-
Min.:	-9999	Unit:	rpm
Max.:	9999	Data Type:	Int16
Default:	0	Change:	Unchangeable

Value Range:

-9999rpm to 9999rpm

Description

Indicates the round actual motor speed, which is accurate to 1 rpm.

Set in H0A.25 (Filter time constant of speed feedback display) the filter time constant for H0b.00.

H0b.01 **Speed reference**

Hexadecimal:	200b-02h	Effective Time:	-
Min.:	-9999	Unit:	rpm
Max.:	9999	Data Type:	Int16
Default:	0	Change:	Unchangeable

Value Range:

-9999rpm to 9999rpm

Description

Indicates the present speed reference (accurate to 1rpm) of the drive in the position and speed control modes.

H0b.02 **Internal torque reference**

Hexadecimal:	200b-03h	Effective Time:	-

Min.:	-300.0	Unit:	%
Max.:	300.0	Data Type:	Int16
Default:	0.0	Change:	Unchangeable

Value Range:

-300.0% to 300.0%

Description

Displays present torque reference (accurate to 0.1%). The value 100.0% corresponds to the rated torque of the motor.

H0b.03 Monitored DI status

Hexadecimal:	200b-04h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

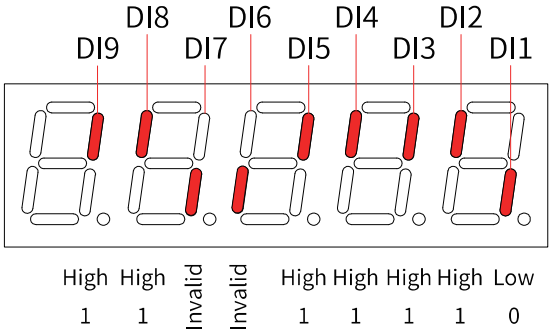
0–65535

Description

Displays the level status of 8 DI terminals without filtering.

Upper LED segments ON: high level (indicated by "1") Lower LED segments ON: low level (indicated by "0")

Assume that the DI1 terminal is low level and DI2 to DI9 terminals are high level, and the corresponding binary number is "110011110". In this case, the value of H0b.03 (Monitored DO signal) read by the software tool is 414 (decimal). See the following figure.



H0b.05 Monitored DO status

Hexadecimal:	200b-06h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

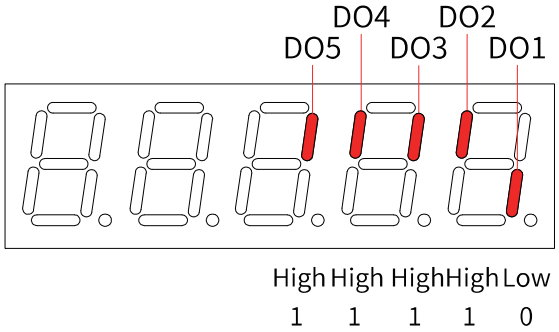
0–65535

Description

Displays the level status of 5 DO terminals without filtering.

Upper LED segments ON: high level (indicated by "1") Lower LED segments ON: low level (indicated by "0")

Assume that the DO1 terminal is low level and DO2 to DO5 terminals are high level, and the corresponding binary number is "11110". In this case, the value of H0b.05 (Monitored DO signal) read by the software tool is 30 (decimal). See the following figure.



H0b.07 Absolute position counter

Hexadecimal:	200b-08h	Effective Time:	-
Min.:	-2147483648	Unit:	Reference unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483648 to 2147483647

Description

Indicates present absolute position (reference unit) of the motor in the position control mode.

This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

H0b.09 Mechanical angle

Hexadecimal:	200b-0Ah	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

Displays present mechanical angle (encoder unit) of the motor. The setpoint 0 indicates the mechanical angle is 0°.
Actual mechanical angle = 360° x H0b.09/(Maximum value of H0b.09 + 1)

Maximum value of H0b.09 for an absolute encoder: 65535

H0b.10 Electrical angle

Hexadecimal:	200b-0Bh	Effective Time:	-
Min.:	0.0	Unit:	°
Max.:	360.0	Data Type:	UInt16
Default:	0.0	Change:	Unchangeable

Value Range:
0.0° to 360.0°

Description
Indicates the present electrical angle of the motor, which is accurate to 0.1°. The electrical angle variation range is ±360.0° during rotation. If the motor has four pairs of poles, each revolution generates four rounds of angle change from 0° to 359°. Similarly, if the motor has five pairs of poles, each revolution generates five rounds of angle change from 0° to 359°.

H0b.11 Speed corresponding to the input position reference

Hexadecimal:	200b-0Ch	Effective Time:	-
Min.:	-9999	Unit:	rpm
Max.:	9999	Data Type:	Int16
Default:	0	Change:	Unchangeable

Value Range:
-9999rpm to 9999rpm

Description
-

H0b.12 Average load rate

Hexadecimal:	200b-0Dh	Effective Time:	-
Min.:	0.0	Unit:	%
Max.:	6553.5	Data Type:	UInt16
Default:	0.0	Change:	Unchangeable

Value Range:
0.0% to 6553.5%

Description

Displays the percentage of the average load torque to the rated torque of the motor, which is accurate to 0.1%. The value 100.0% corresponds to the rated torque of the motor.

H0b.13 Input position reference counter

Hexadecimal:	200b-0Eh	Effective Time:	-
Min.:	-2147483648	Unit:	Reference unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483648 to 2147483647

Description

-

H0b.15 Encoder position deviation counter

Hexadecimal:	200b-10h	Effective Time:	-
Min.:	-2147483648	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483648 to 2147483647

Description

-

H0b.17 Feedback pulse counter

Hexadecimal:	200b-12h	Effective Time:	-
Min.:	-2147483648	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483648 to 2147483647

Description

Used to count the position pulses fed back by the encoder in any control mode. This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

H0b.19 Total power-on time

Hexadecimal:	200b-14h	Effective Time:	-
Min.:	0.0	Unit:	s
Max.:	214748364.7	Data Type:	UInt32
Default:	0.0	Change:	Unchangeable

Value Range:

0.0s–214748364.7s

Description

Used to record the total operating time of the servo drive.

This parameter is a 32-bit integer, which is displayed as a decimal on the keypad.

Note:

If the servo drive is switched on and off repeatedly within a short period of time, a deviation within 1h may be present in the total power-on time record.

H0b.24 RMS value of phase current

Hexadecimal:	200b-19h	Effective Time:	-
Min.:	0.00	Unit:	A
Max.:	655.35	Data Type:	UInt16
Default:	0.00	Change:	Unchangeable

Value Range:

0.00 A to 655.35 A

Description

Displays the RMS value of the phase current of the motor, accurate to 0.01 A.

H0b.26 Bus voltage

Hexadecimal:	200b-1Bh	Effective Time:	-
Min.:	0.0	Unit:	V
Max.:	6553.5	Data Type:	UInt16
Default:	0.0	Change:	Unchangeable

Value Range:

0.0 V to 6553.5 V

Description

Displays the DC bus voltage of the main circuit input voltage after rectification, which is accurate to 0.01 V.

H0b.27 Module temperature

Hexadecimal:	200b-1Ch	Effective Time:	-

Min.:	0	Unit:	°C
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0°C to 65535°C

Description

Indicates the temperature of the module inside the servo drive, which can be used as a reference for estimating the actual temperature of the drive.

H0b.28 Absolute encoder fault information given by FPGA

Hexadecimal:	200b-1Dh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.29 System status information given by FPGA

Hexadecimal:	200b-1Eh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.30 System fault information given by FPGA

Hexadecimal:	200b-1Fh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.33 Fault log

Hexadecimal:	200b-22h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	19	Data Type:	UInt16
Default:	0	Change:	Immediately

Value Range:

0: Present fault

1: Last fault2: 2nd to last fault3: 3rd to last fault4: 4th to last fault

5: 5th to last fault 6: 6th to last fault7: 7th to last fault8: 8th to last fault9: 9th to last fault10: 10th to last fault11: 11th to last fault12: 12th to last fault13: 13th to last fault14: 14th to last fault15: 15th to last fault16: 16th to last fault17: 17th to last fault18: 18th to last fault19: 19th to last fault

Description

Used to view the latest 20 faults of the drive.

H0b.34 Fault code of the selected fault

Hexadecimal:	200b-23h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.35 Time stamp upon occurrence of the selected fault

Hexadecimal:	200b-24h	Effective Time:	-
Min.:	0.0	Unit:	s
Max.:	214748364.7	Data Type:	UInt32
Default:	0.0	Change:	Unchangeable

Value Range:

0.0s–214748364.7s

Description

-

H0b.37 Motor speed upon occurrence of the selected fault

Hexadecimal:	200b-26h	Effective Time:	-
Min.:	-32767	Unit:	rpm

Max.:	32767	Data Type:	Int16
Default:	0	Change:	Unchangeable

Value Range:

-32767rpm to 32767rpm

Description

-

H0b.38 Motor phase U current upon occurrence of the selected fault

Hexadecimal:	200b-27h	Effective Time:	-
Min.:	-327.67	Unit:	A
Max.:	327.67	Data Type:	Int16
Default:	0.00	Change:	Unchangeable

Value Range:

-327.67 A to 327.67 A

Description

-

H0b.39 Motor phase V current upon occurrence of the selected fault

Hexadecimal:	200b-28h	Effective Time:	-
Min.:	-327.67	Unit:	A
Max.:	327.67	Data Type:	Int16
Default:	0.00	Change:	Unchangeable

Value Range:

-327.67 A to 327.67 A

Description

-

H0b.40 Bus voltage upon occurrence of the selected fault

Hexadecimal:	200b-29h	Effective Time:	-
Min.:	0.0	Unit:	V
Max.:	6553.5	Data Type:	UInt16
Default:	0.0	Change:	Unchangeable

Value Range:

0.0 V to 6553.5 V

Description

-

H0b.41 DI status upon occurrence of the selected fault

Hexadecimal:	200b-2Ah	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.42 DO status upon occurrence of the selected fault

Hexadecimal:	200b-2Bh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.43 Group No. of the abnormal parameter

Hexadecimal:	200b-2Ch	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.44 Offset of the abnormal parameter within the parameter group

Hexadecimal:	200b-2Dh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.45 Internal fault code

Hexadecimal:	200b-2Eh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.46 Absolute encoder fault information given by FPGA upon occurrence of the selected fault

Hexadecimal:	200b-2Fh	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.47 System status information given by FPGA upon occurrence of the selected fault

Hexadecimal:	200b-30h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.48 System fault information given by FPGA upon occurrence of the selected fault

Hexadecimal:	200b-31h	Effective Time:	-

Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.51 Internal fault code upon occurrence of the selected fault

Hexadecimal:	200b-34h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.52 Timeout fault flat bit given by FPGA upon occurrence of the selected fault

Hexadecimal:	200b-35h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:

0 to 65535

Description

-

H0b.53 Position deviation counter

Hexadecimal:	200b-36h	Effective Time:	-
Min.:	-2147483648	Unit:	Reference unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483648 to 2147483647

Description

-

H0b.55 Motor speed actual value

Hexadecimal:	200b-38h	Effective Time:	-
Min.:	-6000.0	Unit:	rpm
Max.:	6000.0	Data Type:	Int32
Default:	0.0	Change:	Unchangeable

Value Range:

-6000.0rpm to 6000.0rpm

Description

Indicates the round actual motor speed, which is accurate to 1 rpm.

Set in H0A.25 (Filter time constant of speed feedback display) the filter time constant for H0b.00.

H0b.57 Bus voltage of the control circuit

Hexadecimal:	200b-3Ah	Effective Time:	-
Min.:	0.0	Unit:	V
Max.:	65535.0	Data Type:	UInt16
Default:	0.0	Change:	Unchangeable

Value Range:

0.0 V to 65535.0 V

Description

Displays the bus voltage of the control circuit.

H0b.58 Mechanical absolute position (low 32 bits)

Hexadecimal:	200b-3Bh	Effective Time:	-
Min.:	-2147483647	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483647 to 2147483647

Description

Displays the low 32-bit value (encoder unit) of the mechanical position feedback when the absolute encoder is used.

H0b.60 Mechanical absolute position (high 32 bits)

Hexadecimal:	200b-3Dh	Effective Time:	-
Min.:	-2147483647	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:
-2147483647 to 2147483647
Description
Displays the high 32-bit value (encoder unit) of the mechanical position feedback when the absolute encoder is used.

H0b.64 Real-time input position reference counter

Hexadecimal:	200b-41h	Effective Time:	-
Min.:	-2147483648	Unit:	Reference unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:
-2147483648 to 2147483647
Description
Displays the value of the pulse reference counter before being divided or multiplied by the electronic gear ratio. This value is independent of the servo drive status and the control mode.

H0b.63 NotRdy state

Hexadecimal:	200b-22h	Effective Time:	-
Min.:	0	Unit:	-
Max.:	7	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:
1: Control circuit error
2: Main circuit power input error
3: Bus undervoltage
4: Soft start failed
5: Encoder initialization undone
6: Short circuit to ground failed
7: Others
Description
-

H0b.66 Encoder temperature

Hexadecimal:	200b-43h	Effective Time:	-
Min.:	-32768	Unit:	°C
Max.:	32767	Data Type:	Int16
Default:	0	Change:	Unchangeable

Value Range:
-32768°C to 32767°C
Description
-

H0b.70 Number of revolutions recorded in the absolute encoder

Hexadecimal:	200b-47h	Effective Time:	-
Min.:	0	Unit:	Rev
Max.:	65535	Data Type:	UInt16
Default:	0	Change:	Unchangeable

Value Range:
0Rev–65535Rev
Description
-

H0b.71 Single-turn position fed back by the absolute encoder

Hexadecimal:	200b-48h	Effective Time:	-
Min.:	0	Unit:	Encoder unit
Max.:	2147483647	Data Type:	UInt32
Default:	0	Change:	Unchangeable

Value Range:
0 to 2147483647
Description
Displays the position feedback of the absolute encoder within one turn.

H0b.73 Single-turn offset position of absolute encoder

Hexadecimal:	200b-4Ah	Effective Time:	-
Min.:	0	Unit:	Encoder unit
Max.:	2147483647	Data Type:	UInt32
Default:	0	Change:	Unchangeable

Value Range:
0 to 2147483647
Description
-

H0b.75 Load inertia ratio in online inertia auto-tuning

Hexadecimal:	200b-4Ch	Effective Time:	-

Min.:	0.00	Unit:	-
Max.:	655.35	Data Type:	UInt16
Default:	0.00	Change:	Unchangeable

Value Range:

0.00 to 655.35

Description

-

H0b.76 External load in online inertia auto-tuning

Hexadecimal:	200b-4Dh	Effective Time:	-
Min.:	0.0	Unit:	-
Max.:	6553.5	Data Type:	UInt16
Default:	0.0	Change:	Unchangeable

Value Range:

0.0 to 6553.5

Description

-

H0b.77 Absolute position fed back by the absolute encoder (low 32 bits)

Hexadecimal:	200b-4Eh	Effective Time:	-
Min.:	-2147483647	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483647 to 2147483647

Description

-

H0b.79 Absolute position fed back by the absolute encoder (high 32 bits)

Hexadecimal:	200b-50h	Effective Time:	-
Min.:	-2147483647	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:

-2147483647 to 2147483647

Description

-

H0b.81 **Load position within one turn in absolute position rotation mode (low 32 bits)**

Hexadecimal:	200b-52h	Effective Time:	-
Min.:	-2147483647	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:
-2147483647 to 2147483647
Description
-

H0b.83 **Load position within one turn in absolute position rotation mode (high 32 bits)**

Hexadecimal:	200b-54h	Effective Time:	-
Min.:	-2147483647	Unit:	Encoder unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:
-2147483647 to 2147483647
Description
-

H0b.85 **Load position within one turn in absolute position rotation mode**

Hexadecimal:	200b-56h	Effective Time:	-
Min.:	-2147483647	Unit:	Reference unit
Max.:	2147483647	Data Type:	Int32
Default:	0	Change:	Unchangeable

Value Range:
-2147483647 to 2147483647
Description
-

4.13 H0C Communication Parameters

H0C.00 **Drive axis address**

Hexadecimal:	200C-01h	Effective Time:	Real time
Min.:	0	Unit:	-